

IMO Training 2020-21 (Phase 1)

Level 1

Combinatorial Geometry

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1 Counting

Theorem 1.1 An n -sided polygon has $\frac{1}{2}(n^2 - 3n)$ diagonals.

Theorem 1.2 n lines in a plane divide the plane into at most $\frac{1}{2}(n^2 + n + 2)$ regions.

Theorem 1.3 n planes in a space divide the space into at most $\frac{1}{6}(n^3 + 5n + 6)$ regions.

Problem 0 n points on the circumference of a circle are joined pairwise by line segments. Find the maximum number of regions into which the line segments divide the circle.

Theorem 1.4 (Euler's formula) If a convex polyhedron has V vertices, E edges and F faces, then

$$V - E + F = 2.$$

2 Convex Hulls

Definition 2.1 A set S of points in a plane is *convex* if for any points A, B in S , every point on the line segment joining A and B is also in S .

Theorem 2.2 An intersection of convex sets is a convex set.

Definition 2.3 The convex hull of a set X of points in a plane is the intersection of all convex sets containing X .

Theorem 2.4 The convex hull of a finite set of 3 or more points in a plane is a convex polygon.

Problem 1 Given 4 points in a plane, not all on the same line, prove that at least one of the triangles formed by these points is not acute.

Problem 2 Suppose there are $n > 3$ points in a plane such that no three of them are collinear and every four of them are the vertices of a convex quadrilateral. Prove that the n points are the vertices of a convex n -sided polygon.

Problem 3 (IMO 2011) Let $\mathcal{S}$ be a finite set of at least two points in the plane. Assume that no three points of $\mathcal{S}$ are collinear. A *windmill* is a process that starts with a line ℓ going through a single point $P \in \mathcal{S}$. The line rotates clockwise about the pivot P until the first time that the line meets some other point belonging to $\mathcal{S}$. This point, Q , takes over as the new pivot, and the line now rotates clockwise about Q , until it next meets a point of $\mathcal{S}$. This process continues indefinitely.

Show that we can choose a point P in $\mathcal{S}$ and a line ℓ going through P such that the resulting windmill uses each point of $\mathcal{S}$ as a pivot infinitely many times.

3 Coloring

Problem 4 (CMO 1986) Each point in a plane is colored black or white arbitrarily. Prove that there exists an equilateral triangle of side length 1 or $\sqrt{3}$ whose three vertices have the same color.

4 Extremal Principle

Theorem 4.1 Any finite set of real numbers contains a smallest element and a largest element.

Problem 5 Suppose there are n red points and n blue points in a plane. Among these $2n$ points, no three are collinear. Prove that there exist n line segments, each joining a red point and a blue point, such that no two of the line segments intersect.

Problem 6 (Sylvester-Gallai theorem) Given a finite set of points in a plane, not all collinear, prove that there exists a line that passes through exactly two of the points.